Geophys. J. R. astr. Soc. (1982) **69**, 573–578

Letter to the editors

Comment on 'A palaeomagnetic study of Cambrian red beds from Carteret, Normandy, France' by W. A. Morris

H. Perroud and N. Bonhommet *Laboratoire de Géophysique Interne, CAESS (CNRS), Université de Rennes, 35042 – Rennes Cédex, France*

M. Robardet *Laboratoire de Stratigraphie, CAESS (CNRS), Université de Rennes, 35042 – Rennes Cédex, France*

Received 1981 September 25

Downloaded from https://academic.oup.com/gji/article/69/2/573/767272 by Curtin University Library user on 26 May 2026

Conclusions of 'A palaeomagnetic study of Cambrian red beds from Carteret, Normandy, France' (Morris 1980) have already been reviewed by Van der Voo & Morris (1982). These two authors corrected some errors which occurred in the original publication and concluded that the two pole positions obtained from these Cambrian sediments 'match very convincingly the apparent polar wander path for the Eocambrian and Cambrian constructed for the Armorican Massif of France by Hagstrum, Van der Voo, Auvray & Bonhommet (1980)'.

As it has already been stressed by Van der Voo & Morris, an Ordovician age of the A magnetization postdating a Middle to Upper Cambrian folding cannot be accepted. Research carried out in the northern part of the Armorican Massif has demonstrated that all the Palaeozoic structural deformations took place at the end of Palaeozoic times, the earliest ones being of Upper Devonian to Lower Carboniferous age (i.e. the so-called Bretonic phase). In Normandy the Lower Ordovician (Arenig) transgression (Grès armoricain Formation) follows a stratigraphical gap ('lacune normande') as the sedimentation stopped in the Cotentin peninsula during Cambrian times but no structural event of Cambrian age has ever been demonstrated (Doré 1972; Robardet 1973). In addition to this statement, we consider that a careful study of the original arguments of Morris (1980) reveals numerous other problems concerning the geology of the investigated area, the interpretation of alternating field, thermal and chemical demagnetization results, the proposed ($A \rightarrow B$) sequence of remanence acquisition and the age of the shallow B component which we interpret as a Hercynian remagnetization.

1 Geology

1.1 STRATIGRAPHY

Minor inaccuracies may be pointed out such as the precise stratigraphical position of the hyolitids and sponge spicules which are not known from the red beds at Dennemont Quarry as stated by Morris (1980, p. 578) but from the underlying argillaceous sandstones with calcareous nodules (Doré 1963). Moreover these red beds are only 50 m thick whereas the simplified lithological column drawn by Morris (Fig. 1, p. 580) after Doré (1972) (this

paper being omitted in Morris' references) shows these beds as being 500 m thick. Much more important are the correlations proposed by Morris between the Cambrian successions at Carteret and le Rozel. He considers (p. 578) that 'the Pointe du Rozel strata are stratigraphically higher than the Dennemont Quarry section'. As a matter of fact, both successions correspond to the same sedimentary cycle of Lower Cambrian age but precise correlations between the two are impossible owing to the lack of fossils at le Rozel and to the monotonous lithology in this locality where different lithological units cannot be distinguished as they are at Carteret (cf. Doré 1972). Considering the southern limb of the Siouville syncline where several hundred metres of Cambrian sediments are overlying the Pointe de Rozel strata (the sample location at le Rozel being incorrectly placed too high in the sequence by Morris (Fig. 1)) it could well be argued that these strata are older than those at Carteret. Any argumentation based on an hypothetical correlation between the two sequences (cf. Morris p. 581) would remain highly questionable.

1.2 STRUCTURE

The two sampled localities of le Rozel and Carteret are located in two different structural units, namely the Siouville syncline and the Les Moitiers-d'Allonne anticline. The structural relations between these two ENE–WSW trending units might not be as simple as it could be supposed, due to the occurrence of major faults between them.

Moreover, the Cambrian strata cropping out at Carteret along the coast (cape and beach of Carteret) and in Dennemont Quarry have an anomalous almost N–S trend (N 330° to 340° with 20° to the ENE dip; cf. Table 1) as already emphasized by Doré (1963, p. 724). The dip corrections made by Morris who read Doré's (1963) paper (see Morris references) show that he used wrong data, trend and dip having been confused. The N–S trend of Carteret strata in the ENE–WSW (N 60°) trending Les Moitiers-d'Allonne anticline make them appear as an anomaly and one can wonder how to restitute their primary structure.

Table 1. Mean-site magnetization direction for the two components *A* and *B* before (IS) and after (DC) tilt correction.

		A				*B*				Structural position	
		IS		DC		IS		DC			
		D	*I*	*D*	*I*	*D*	*I*	*D*	*I*	Strike	Dip
Carteret	4	261	69	292	81						
	3	261	62	278	74						
	8	258	78	5	86						
	7	240	73	240	87						
	2	272	68	307	78	204	32	197	43		
	1					213	30	208	42		
	5	281	78	2	81	237	28	236	42		
	6	298	75	355	76	209	18	206	30		
	Mean	266	72.5	316	82	216	28	211	40	330	14
		$k = 105$		$k = 105$		$k = 32$		$k = 33$		$k = 1255$	
		$\alpha = 6°$		$\alpha = 6°$		$\alpha = 14°$		$\alpha = 14°$		$\alpha = 1.5, N = 9$	
Rozel	9					203	07	205	4	196	23
	10	226	83	285	54	205	−5	202	−4	206	33
	11	174	64	231	64	200	−2	199	0	203	26
	Mean	185	75	262	62	203	0	202	0		
		$k = 26$		$k = 16$		$k = 145$		$k = 263$			
						$\alpha = 7°$		$\alpha = 6°$			

Downloaded from https://academic.oup.com/gji/article/69/2/573/767272 by Curtin University Library user on 26 May 2026

1.3 METAMORPHISM

The granite massif of Flamanville, which intruded the southern part of the Siouville syncline during the Carboniferous is located close to (4 km) and north of le Rozel. This batholith caused a thermometamorphic aureol to form in the Palaeozoic formations which include Cambrian strata immediately to the north of le Rozel. In the Pointe du Rozel several basic dykes are known which might be associated with the granitic magmatism, as well as a micro-granitic dyke a few tens of metres from sites 9 and 11 sampled by Morris which is identical to those crossing the granite massif at Flamanville. All this clearly shows that the Rozel Cambrian strata are located in the influence area of the granite massif which was studied palaeomagnetically some years ago (Van der Voo & Klootwijk 1972). It would consequently be most unlikely to find at le Rozel a main remanence direction ('FB') which (according to Morris, p. 587) 'definitely predates folding' and 'probably dates from the initial depositional event'.

2 NRM behaviour upon cleaning

Behaviour during af treatment as illustrated in Morris (fig. 2, p. 580) may seem very surprising for red beds where haematite pigment is the main carrier of remanence. For the three sites (2, 6, 10) which show the most significant variation after af (tables 1 and 2 in Morris's paper, pp. 579–581), Fig. 1 shows plots of mean site directions given by Morris before and after cleaning without tilt correction. The evolution of vectors towards shallow inclinations is clearly due to the elimination of a recent component as great circles of magnetization are very close to the present direction of the magnetic field at Carteret. At site 10 this component was dominant as only 10 per cent of magnetization remains after treatment (Morris, p. 581). It would be unusual to find that af demagnetization of red beds could discriminate between two ancient components; as tilt corrections were made with incorrect field data (see above) Morris' fig. 2 (p. 508) cannot be used to show the trend towards shallow inclination as evidence for a primary B component and the sequence of acquisition to be $B \rightarrow A$.

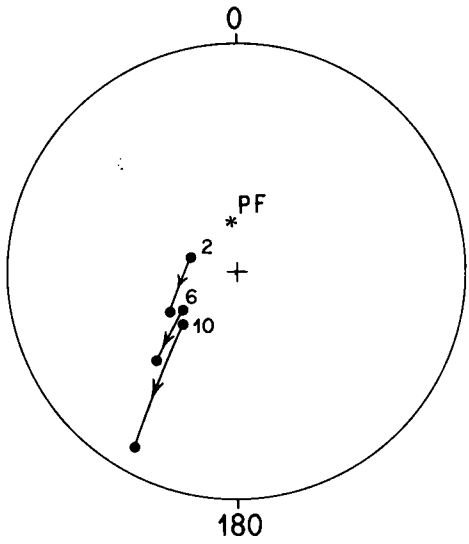


Figure 1. Evolution of NRM site means (sites, 2 6 and 10) during af treatment (*in situ* directions). PF is the present field at Carteret.

Downloaded from https://academic.oup.com/gji/article/69/2/573/767272 by Curtin University Library user on 26 May 2026

Following af demagnetization, heating of the same speciment does not lead to variation of direction but as stated by Morris (p. 583): 'further thermal treatment produced a shift towards steep inclinations. Acid treatment was then commenced and this was associated with a marked increase of remanence intensity ...'. The examples given by Morris (fig. 4a, b) clearly show that a steep component appeared at high temperatures; it seems wise not to use data given by leaching because of the strange increase in the intensity (was it done in zero field?) contrary to Morris' interpretation of a contamination due to heating. Acid leaching was then used with success after af on some specimens to separate the shallow component B. Our experience with red beds (Perroud 1980) is that the components may not necessarily appear in the same order with acid leaching as with thermal demagnetization.

In conclusion, from cleaning experiments made by Morris, af demagnetization shows a recent component (present Earth field) existing at least in sites 2, 6 and 10, which can be dominant (site 10); component A with steep inclination corresponds to high blocking temperatures and a component B with shallow inclination exists which is more easily isolated with af demagnetization followed by acid leaching as shown by Morris.

3 Sequence of remanence acquisition

The $B \rightarrow A$ sequence (shallow to steep) proposed by Morris is based on three types of argument. (1) The evolution during af: we have seen that it is due to the existence of a soft component acquired in the present Earth field. Evidence from thermal treatment would be in favour of the reverse order as A corresponds to high blocking temperatures. (2) A fold test on the two sites of Carteret and Rozel: this test is not significant as tilt correction was made using incorrect data. Moreover Carteret and Rozel do not belong to the same tectonic unit (see geology). (3) Petrology: the photographs (plate 1 in Morris's paper) show in fact a variable haematite content which corresponds simply to different facies and colours. Our own field observations would suggest a secondary origin for the pigment at le Rozel where red spots are crossing the stratification.

Contrary to Morris' proposition, we consider that correct sequence is $A \rightarrow B$ because of the following arguments. (1) Using Morris' data we find (Table 1) that component A at Carteret without tilt correction is different from all post-Hercynian directions and therefore must be pre-Carboniferous which means that a tilt correction is necessary; our own structural directions lead to $D = 316°$, $I = +82°$, $\alpha_{95} = 6°$. But we do not make any fold test with the data from Rozel, because we do not feel that it is justified. (Morris' data are not given in the paper but can be recovered by plotting on a stereonet nrm directions before and after tilt correction as given by Morris in table 1, p. 583.) (2) Component B at Rozel is the same before and after tilt correction; since the magnetization is close to the strike, a fold test between Rozel sites is not significant, see Table 1. (3) The direction $D = 203, I = 0$ is a very common Hercynian direction in the Armorican Massif (for example, the Hercynian granites of Flamanville, (Van der Voo & Klootwijk 1972), and of Ploumanach (Duff 1979). Therefore, noting that le Rozel is close to Flamanville (see Section1) it is wise to regard the B direction as Carboniferous and not as Lower Cambrian.

Directions given in Table 1 for the A component at le Rozel (site 11) and the B component at Carteret (sites 1, 2, 5 and 6) are interpreted as vectorial sums of A and B which have not been separated by the demagnetization treatments. Fig. 2 shows that, after tilt correction these directions are close to the great circle A–B, a result of the pretectonic nature of A, and the invariance of B with tilt correction. The A component in site 10 (Table 1) is probably the result of a B component superposed on a recent component; because of the low coercivities involved it would be surprising to find a third direction of Cambrian age.

Downloaded from https://academic.oup.com/gji/article/69/2/573/767272 by Curtin University Library user on 26 May 2026

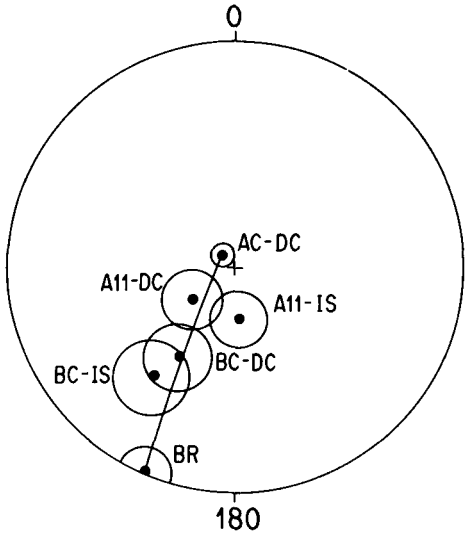


Figure 2. Site mean directions with their circle of confidence. *A*, *B* are the names of the components in Morris' paper. C is for Carteret, R for le Rozel. 11 is for site 11, IS and DC states for *in situ* or dip corrected, respectively.

We therefore interpret *A* as contemporaneous with deposition or diagenesis and *B* as a remagnetization during Hercynian times.

4 Conclusions and palaeomagnetic poles

Using Morris' data we propose an interpretation which differs from those published by Morris, and Van der Voo & Morris, as we consider the sequence of remanence acquisition to be $A \rightarrow B$ (steep to shallow) and not $B \rightarrow A$. We show the *B* component at le Rozel to be secondary and Carboniferous, contemporaneous of the local intrusion of the Flamanville (300 Myr) granite; the pole position located at 341°E, 37°S (BR in Fig. 3) fits well with other

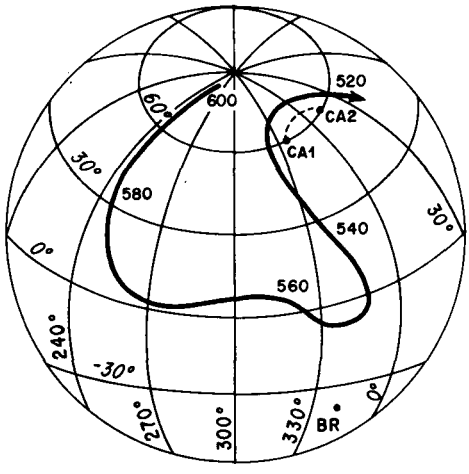


Figure 3. Pole positions BR (component *B*, le Rozel), CA1 and CA2 (component *A*, Carteret, see text) compared with the Eocambrian–Cambrian APWP for the Armorican Massif (Hagstrum *et al.* 1980).

Downloaded from https://academic.oup.com/gji/article/69/2/573/767272 by Curtin University Library user on 26 May 2026

Carboniferous poles from the Armorican Massif. We argue that the *A* component at Carteret is primary; the corresponding pole position (CA1 in Fig. 3) is located at 59°N, 331°E, which is very close to Duff results on the Cambrian Jersey volcanics (522 Myr in Duff 1980) and is in good agreement with APWP of the Armorican Massif (Hagstrum *et al.* 1980) For Eocambrian–Cambrian times which is not the case with our pole BR.

To regard the CA pole as definitive it would be necessary to take into account the anomalous structural position of Carteret in the Moitiers-d'Allonne anticline (Doré 1963; Robardet 1970) which may have a tectonic origin. A simple correction (a rotation of 80°) has therefore been applied; but because of the very steep inclination, the palaeopole is very close to the site and even after a large rotation the new position (CA2 in Fig. 3) is not very different. However the study of sites consistent with the main structural direction would be necessary to discriminate between the two alternative pole positions.

Acknowledgments

We thank Dr R. Van der Voo for his communication in 1980 December of the manuscript (Van der Voo & Morris 1982) about a first re-evaluation of the data given by Morris in the original paper (Morris 1980).

References

Doré, F., 1963. Age et position des calcaires de Saint-Jean-de-la-Rivière dans le Cambrien des Moitiers d'Allonne (Manche), *Bull. Soc. géol. Fr.*, **5**, 722–729.

Doré, F., 1972. La transgression majeure du Paléozoique inférieur dans le Nord-Est du Massif Armoricain, *Bull. Soc. géol. Fr.*, **14**, 79–93.

Duff, B. A., 1979. The palaeomagnetism of Cambro-Ordovician red beds, the Erquy Spilite Series and the Trégastel-Ploumanac'h granite complex, Armorican Massif (France), *Geophys. J. R. astr. Soc.*, **59**, 345–365.

Duff, B. A., 1980. The palaeomagnetism of Jersey volcanics and dykes, and the Lower Palaezoic apparent polar wander path for Europe, *Geophys. J. R. astr. Soc.*, **60**, 355–375.

Hagstrum, J. T., Van der Voo, R., Auvray, B. & Bonhommet, N., 1980. Eocambrian-Cambrian palaeomagnetism of the Armorican Massif, France, *Geophys. J. R. astr. Soc.*, **61**, 489–517.

Morris, W. A., 1980. A palaeomagnetic study of Cambrian red beds from Carteret, Normandy, France, *Geophys. J. R. astr. Soc.*, **62**, 577–590.

Perroud, H., 1980. Contribution à l'étude paléomagnétique de l'arc Ibéro-Armoricain, *thèse 3ème cycle*, University of Rennes, and *Bull. Soc. geol. Minér. Bretagne*, **14**, 1–100, 1982.

Robardet, M., 1970. Directions caledonienne et armoricaine dans l'anticlinal des Moitiers d'Allone (Manche), *Bull. Soc. linn. Normandie*, **101**, 36–44.

Robardet, M., 1973. Evolution géodynamique du Nord-Est du Massif Armoricain au Paléozoique, *thèse*, University of Paris, and *Mém. Soc geol. Minér. Bretagne*, **20**, 1981.

Van der Voo, R. & Klootwijk, C. T., 1972. Paleomagnetic reconnaissance study of the Flamanville granite, with special reference to the anisotropy of its susceptibility. *Geologie Mijnb.*, **51**, 609–617.

Van der Voo, R. & Morris, W. A., 1982. A short note on early Cambrian palaeomagnetism from Normandy, France, *Geophys. J. R. astr. Soc.*, **68**, 269–271.

Downloaded from https://academic.oup.com/gji/article/69/2/573/767272 by Curtin University Library user on 26 May 2026